Robot dance studio

Build it. Attach it. Dance with the cone.

Today’s robot challenge

A dance that pushes an upright cone

A repeat count you can explain

Keep the cone in its guide through every turn

GREEN steady: on **ORANGE flashing: charge** **RED flashing: stop + mentor**

Center button lights. A letter on the display is your program’s output.

Counted loops and indentation

```
for beat in range(3):
    await drive(90)
    await turn(90)
await signal("D")
```

Our prediction

How many drives happen in three repeats?

Does a signal outside the loop repeat?

Pusher attached; cone in guide. Use the complete supplied project.

Our robot and build jobs

Robot team: predict, run, count. Rotate jobs.

Build team: make a removable front cone pusher.

Attach your pusher. Test the dance. Improve one feature.

Your pusher stays attached during moves AND turns.

Robot turns: rotate every 4 minutes. Retests: every 2 minutes.

Build and attach your cone pusher

1. Match two front mounting points. Build using the proxy.
2. Add a low bar, short side guides and a brace.
3. Hand-check a gentle push AND turn. Keep wheels clear.
4. Attach YOUR pusher to the stopped robot. Check the fit.
5. Run the dance with the cone. Improve one part. Retest.

Success: 3 repeats. Secure mount. Cone upright inside the guide.

Rotate jobs. A hand test alone does not finish this challenge.

Evidence from our test

```
for beat in range(3):
    await drive(90)
    await turn(90)
await signal("D")
```

Pusher stays attached.
Cone stays in guide.
Count full repeats: 0–3.

Predict: drives, turns, signals?

What did our design change improve?

Stop for a tip, escape or loose mount. Reset only when stopped. SPEED = 200.

What we learned and noticed

What did you learn about repeating actions?

What did you notice during the robot dance?

What would you change in the loop or cone pusher?

Save your team project. Store your build. Reset the robot and parts.